

ABDUR REHMAN AFZAL

Senior Backend Python Engineer | Full-Stack Developer | AI Engineer

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PROFESSIONAL SUMMARY

Senior Backend Python Engineer and Full Stack Python Developer with 7+ years shipping production systems across 9 live domains — from a Dubai luxury real estate platform (b1properties.ae) to a multi-tenant SaaS serving enterprise security clients (workfly.app) and a GCP/Kubernetes fintech engine processing UK mortgage data. Expert in Django, Django REST Framework (DRF), FastAPI, Django Channels, Celery, PostgreSQL, Redis, Amazon Web Services (AWS), and LangChain / Retrieval-Augmented Generation (RAG) — with a track record of cutting infrastructure costs by 30%, reducing manual workloads by 70–80%, and leading cross-functional teams of 5 engineers to deliver on schedule. Targeting Backend Engineering, Full Stack Development, and AI Engineering roles in GCC, Europe, and North America.

TECHNICAL SKILLS

Backend	Python, Django 3/4/5, Django REST Framework (DRF), Django Channels, FastAPI, Flask, Celery, Huey, RabbitMQ, GraphQL, RESTful APIs, WebSockets, gRPC, JSON Web Token (JWT), OAuth 2.0, CORS, Asynchronous Programming, Multithreading, Wagtail CMS, Swagger / OpenAPI (drf-yasg)
AI & ML	LangChain, CrewAI, Retrieval-Augmented Generation (RAG) Pipelines, Large Language Models (LLMs), Generative AI (GenAI), AI Agents, Vector Databases (ChromaDB, Pinecone), OpenAI API, GPT-4, Prompt Engineering, Embeddings, Natural Language Processing (NLP), Computer Vision, YOLOv8 (Ultralytics), PyTorch, OpenCV, Tesseract OCR / EasyOCR, n8n Workflow Automation
Frontend	React 18/19, Next.js 16, Angular 15, TypeScript 5, JavaScript, TailwindCSS, Redux Toolkit, RxJS, PrimeNG, Material-UI, HTML5, CSS3
Cloud & DevOps	Amazon Web Services (AWS) — S3, EC2, RDS, ECS, ECR, SES, CloudWatch, Google Cloud Platform (GCP) — BigQuery, Google Kubernetes Engine (GKE), Artifact Registry, Docker, Kubernetes, Continuous Integration / Continuous Deployment (CI/CD), GitHub Actions, Nginx, Gunicorn, PM2, Linux/Bash, SSL/TLS, Infrastructure as Code
Databases	PostgreSQL, MySQL, MongoDB (NoSQL), Redis, Firebase/Firestore, SQLite, BigQuery, Snowflake, SQL, Database Design, Query Optimization
Architecture	Multi-Tenant SaaS, Microservices Architecture, API-First Design, RESTful API Design, Event-Driven Architecture, Distributed Systems, High Availability, Scalable Systems, Low-Latency Architecture, Async Task Queues, Real-Time WebSockets, Caching Strategies, Database Optimization, Role-Based Access Control (RBAC)
Integrations	Stripe, PayPal, Binance, Twilio, SendGrid, MailChimp, Google Maps, Slack, OpenStack, AWS SES, Pillow, BeautifulSoup4
Practices	Agile / Scrum, Test-Driven Development (TDD), PyTest, Integration Testing, Code Review, Technical Mentoring, Cross-Functional Collaboration, Stakeholder Communication, Git, Jira, API Documentation, Deployment Scripting

PROFESSIONAL EXPERIENCE

B1 Properties | Dubai, UAE Jan 2025 – Present

Full-Stack Python Developer

- Delivered a production luxury real estate platform (b1properties.ae) managing 100+ active property listings across 4 categories, backed by 50+ REST API endpoints, 13 Django ORM models, and 20+ normalised database tables — replacing a manual listing management workflow for a Dubai real estate firm.
- Eliminated 100% of N+1 query incidents across all API endpoints by refactoring the Django ORM layer with `select_related` and `prefetch_related` across 13+ models, producing a measured 30% backend API performance improvement confirmed via AWS CloudWatch latency monitoring.
- Achieved zero production downtime since launch by engineering complete infrastructure stack — Gunicorn multi-worker + Nginx with 7 security headers, PM2-clustered Next.js frontend, SSL/TLS via Let's Encrypt, and automated PostgreSQL backups with 7-day retention — delivering a fully hardened Continuous Integration / Continuous Deployment (CI/CD) production environment from scratch.
- Increased client lead qualification rate by 60% by deploying a LangChain + Retrieval-Augmented Generation (RAG) + ChromaDB chatbot — while simultaneously cutting daily manual reporting effort by 70% through an n8n GPT-4 automation pipeline, reclaiming ~2 hours/day of staff time.
- Reduced client handover and onboarding time by ~80% by authoring 5+ deployment artefacts — deploy scripts, Nginx configs, Systemd service files, PM2 ecosystem config, and `DEPLOYMENT_GUIDE.md` — enabling any engineer to reproduce the production environment independently within hours.

Aircord Technologies | Lahore, Pakistan 2024 – 2025

Senior Python Backend & AI Developer

- Reduced cloud infrastructure costs by 30% across 3+ enterprise SaaS platforms by leading the migration from monolithic Django to AWS-hosted FastAPI microservices — decomposing a single-process system into independently scalable services.
- Cut document processing turnaround time by ~70% across 3+ client pipelines by integrating LangChain, CrewAI, and OpenAI API for Large Language Model (LLM)-powered intelligent document triage — replacing a manual review process that previously required 30–60 minutes per batch.
- Improved API throughput by 35% and cut release cycle time by ~50% by implementing Docker + GitHub Actions Continuous Integration / Continuous Deployment (CI/CD) pipelines — reducing manual deployment steps from ~8 to zero.
- Eliminated hours of daily manual operations by architecting n8n automation pipelines for complex multi-step business workflows across 3+ client accounts.

Synares Systems | Lahore, Pakistan 2023 – 2024

Senior Software Engineer II / Backend Team Lead

- Eliminated 50% of workforce admin overhead for security guard companies by leading development of TimeKeepers / WorkFly (workfly.app) — a live multi-tenant SaaS spanning 8 Django apps, 5 user role types, 64+ Angular components, and a Kotlin Android app, replacing entirely manual scheduling and reporting workflows.
- Delivered sub-100ms real-time response latency for live guard operations by architecting a Django Channels + WebSockets + Redis pub/sub event-driven layer — replacing a polling-based system that updated every 15–30 seconds, achieving ~200× improvement in notification delivery speed.
- Reduced payroll discrepancy incidents to near zero by engineering a GPS-based geofencing time-clock system that auto-validates clock-in/out within authorised perimeters — eliminating manual timesheet errors that industry benchmarks show affect 10–15% of weekly entries in manual systems.
- Compressed deployment cycle from hours to under 5 minutes by building a GitHub Actions Continuous Integration / Continuous Deployment (CI/CD) pipeline: Docker build → AWS ECR push → ECS task definition update → production rollout — across two fully isolated DEV and Production environments.
- Scaled engineering output by leading 5 backend engineers, introducing Black, pre-commit hooks, and PyTest coverage standards — reducing code review cycle time by ~40% and cutting post-merge defect rate through automated quality gates on every pull request.

Merik Solutions | Islamabad, Pakistan 2022 – 2023

Full-Stack Python Developer

- Reduced production bug rate by 30% across 3+ enterprise Django applications by introducing PyTest and UnitTest test suites from zero coverage — establishing Test-Driven Development (TDD) as the team engineering standard.

- Delivered Wagtail CMS platforms and DRF API layers for content-heavy enterprise clients, cutting editorial publishing time by replacing manual workflows with structured content management.
- Cut client reporting turnaround from days to minutes by delivering automated operational dashboards, surfacing real-time data previously locked in manually compiled spreadsheets — directly reducing decision latency for 3+ enterprise clients.

Viral Square | Lahore, Pakistan 2021 – 2022

Python Developer

- Built and scaled Palletfly (dbr.palletfly.com) to 40,000+ lines of production Python across 10 Django apps, 91+ ORM models, and 380+ schema migrations — managing the complete B2B supply chain lifecycle for logistics clients handling 10,000+ product SKUs.
- Eliminated inventory sync conflicts for concurrent users by implementing a real-time collaboration engine (Django Channels + WebSockets + Redis pub/sub) enabling zero-conflict simultaneous editing.
- Reduced bulk data operation time by ~90% (from 2–3 hours to under 15 minutes) for Google Sheets imports and Amazon ASIN sync jobs by building a Huey + Redis async task engine — unblocking the request thread and eliminating user-facing timeouts on large dataset operations.
- Extended platform reach to external B2B clients by architecting a decoupled DBR_API microservice — a separate Django/DRF service sharing the production RDS PostgreSQL database via token-authenticated endpoints.

UEnergy Solar | Huddersfield, UK (Remote) 2018 – 2021

Software Engineer

- Engineered and maintained 2 production platforms (uenergysolar.co.uk and uenergysolar.com) supporting a UK enterprise renewable energy company with 20+ named enterprise clients including BP, Shell, NHS, Sainsbury's, ASDA, and 10+ UK local councils — serving domestic and commercial solar markets across the UK and Pakistan.
- Accelerated sales proposal generation by ~60% by delivering a Django + DRF + PostgreSQL CRM and quoting system — enabling the sales team to generate site-specific energy proposals in minutes vs hours and track dozens of concurrent UK installation projects from a single pipeline view.
- Consolidated operations across dozens of concurrent UK energy installations by building a project and site management backend — covering job tracking, engineer scheduling, and multi-site status management in a single Django + PostgreSQL system.
- Eliminated manual billing errors for 20+ enterprise client accounts by implementing an automated invoicing and billing module integrated with the CRM — replacing a spreadsheet-based process that required manual invoice generation and payment reconciliation.
- Delivered real-time ROI visibility to 20+ enterprise clients by building an energy monitoring and analytics dashboard with solar output reporting, cost savings calculations, and Chart.js visualisations — replacing manually compiled monthly PDF reports with an always-current self-service view.

University of Central Punjab (UCP) | Lahore, Pakistan 2018 – 2021

CS Instructor (Part-Time)

- Taught 200+ undergraduate students across 7 Computer Science courses over 3 academic years — OOP, Data Structures & Algorithms (DSA), Operating Systems, Networks, ITC, COAL, and Programming Fundamentals — building the foundational engineering communication skills that underpin technical mentorship today.

KEY PROJECTS

B1 Properties – Luxury Real Estate Platform (Production)

Tech: *Django 5.2, DRF, PostgreSQL, Next.js 16, TypeScript 5, TailwindCSS 4, AWS S3, Nginx, Unicorn, PM2, JWT*

- End-to-end production platform live at b1properties.ae — 13+ Django models, 50+ REST API endpoints, 20+ normalised DB tables, 15+ Next.js pages.
- Full DevOps ownership: Ubuntu VPS setup, Nginx reverse proxy, SSL/TLS, automated DB backups, deployment scripting, and complete client handover documentation.
- Advanced property search with 10+ filter criteria, AWS S3 media storage, Swagger API docs, and appointment booking system with form validation.

Portfolio RAG Assistant – AI Chatbot with Lead Capture & Guardrails (abdurrehmanafzal.cloud)

Tech: FastAPI, OpenAI GPT-4o-mini, ChromaDB, Sentence-Transformers (all-MiniLM-L6-v2), Pydantic, Python

- Built a production-style Retrieval-Augmented Generation (RAG) assistant that answers recruiter questions about my professional background — local sentence-transformers embeddings, ChromaDB persistent vector store, top-k semantic retrieval, and GPT-4o-mini grounded generation with per-session multi-turn conversation memory.
- Implemented OpenAI function/tool calling so the LLM autonomously fetches structured project details from a backing data source, plus Pydantic structured-output lead extraction that auto-detects and captures recruiter contact info (name, email, budget, intent) from free-form chat.
- Engineered input/output guardrails against prompt-injection (keyword + length filtering, system-prompt-leak detection) and a keyword-based RAG evaluation harness (/evaluate endpoint) scoring answer relevance against an expected-keyword test set.

B1 Chatbot – AI Real Estate Assistant (RAG Architecture)

Tech: GPT-4, LangChain, ChromaDB, Django, Vercel, Telegram Bot API

- Production RAG chatbot with vector search, custom embeddings, proprietary knowledge base, and Telegram Bot integration for real-time property enquiries.
- Reduced repetitive agent enquiry handling by 60% through AI-driven automated responses.

AI-Assisted News Intelligence Platform

Tech: Node.js, Express.js, PostgreSQL, Sequelize ORM, OpenAI GPT-4, React, Cloudinary CDN, Nodemailer, node-cron, PDFKit

- Engineered an end-to-end AI pipeline: multi-source RSS ingestion (Google News, Arabian Business, Gulf News, Khaleej Times) → intelligent tag-based filtering (30+ categorised keywords with priority weighting) → GPT-4 summarisation → PDFKit report generation → Cloudinary CDN storage → automated Nodemailer email delivery.
- Designed a dynamic prompt templating engine with variable interpolation (organisation, country, keywords, articles) enabling context-aware, client-specific intelligence summaries across gpt-3.5-turbo and gpt-4 models.
- Built a 12-component React dashboard (6 specialised tabs: Overview, Configuration, AI Prompts, Reports, Real Estate Daily, Tags Management) with React Context-based dark/light theming, real-time status updates, and node-cron-driven scheduled report automation.
- Implemented article deduplication logic with a 7-day rolling window across 4 concurrent RSS feeds, ensuring report freshness without duplicate content processing.

EasyKost – AI-Driven Product Cost Estimation System

Tech: Django, DRF, Firestore, Firebase Storage, gRPC, Microservices

- Migrated monolithic system to microservices architecture (Auth, AI, Frontend) with gRPC inter-service communication.
- Led database migration from MySQL to Firestore (NoSQL), significantly improving scalability and query performance.

AdsGency AI – AI-Powered Advertising Platform | adsgency.ai (Contributor)

Tech: FastAPI, Django, DRF, OpenAI API, AWS, Google Ads API, Meta Marketing API

- Built RESTful backend APIs (FastAPI + Django DRF) powering the AdsGency analytics dashboard — a funded AI SaaS platform serving 150+ businesses with \$50M+ in managed ad spend across Google, Meta, LinkedIn, and YouTube.
- Developed Google Ads and Meta Marketing API integrations enabling omnichannel campaign management and real-time performance data synchronisation from within the platform.
- Implemented OpenAI-powered ad content generation module, producing AI-written ad copy and creatives tailored to audience segments and campaign objectives.

Portfolio Voice Agent – Real-Time Conversational Voice RAG | talk.abdurrehmanafzal.cloud

Tech: Pipecat, WebRTC, Deepgram, OpenAI GPT-4o-mini, Cartesia, Silero VAD, Sentence-Transformers, FastAPI, Docker, GitHub Actions

- Built a real-time, interruptible browser voice agent (Pipecat pipeline: WebRTC → VAD → Deepgram STT → GPT-4o-mini → Cartesia TTS) with barge-in and RAG via OpenAI tool calling for grounded, hallucination-free answers.
- Deployed Dockerized (host-networking WebRTC + Nginx/Traefik HTTPS) with a GitHub Actions CI/CD auto-deploy pipeline; added cost guardrails keeping runtime at ~\$0.

VenueGPS – Location-Based Venue & Events Discovery Platform | develop.venuegps.com

Tech: Django, DRF, PostgreSQL, Redis, Google Maps API, Facebook OAuth

- Architected the full Django REST Framework backend for VenueGPS — a location-based venue and events discovery platform — covering multi-role user authentication (email + Facebook OAuth via django-allauth), business listing management, and city-based geospatial venue search using Google Maps API.
- Built a business claiming and verification system allowing venue owners to claim listings with structured profile management (business type, sub-type, address, contact details) and admin approval workflows.
- Implemented a tiered hashtag featuring system with pricing tiers, status tracking, and featured-until-date scheduling — enabling a monetisation layer for promoted venue content.
- Deployed with Redis caching for geolocation query optimisation and PostgreSQL for relational venue, user, and business data — serving a live multi-city platform.

UEnergy Solar – Enterprise Renewable Energy Platform | uenergysolar.co.uk | uenergysolar.com

Tech: Django, DRF, Python, PostgreSQL, AWS, Chart.js

- Built and maintained two production websites (UK and Pakistan) for an enterprise renewable energy company serving BP, Shell, NHS, Sainsbury's, ASDA, and multiple UK local councils.
- Developed a client CRM and quoting system enabling the sales team to manage leads, generate site-specific energy proposals, and track project pipelines across commercial and residential installations.
- Built a project and site management backend covering installation job tracking, engineer scheduling, and multi-site status management across dozens of concurrent UK energy projects.
- Implemented invoicing and billing module integrated with the CRM — automating invoice generation and payment tracking for commercial clients.
- Delivered an energy monitoring and analytics dashboard with solar output reporting, cost savings calculations, and Chart.js visualisations providing clients real-time system performance and ROI insight.

EID Document Processing System

Tech: Django 5.1.2, Python 3.8+, YOLOv8 (Ultralytics), PyTorch 2.4, OpenCV 4.10, Tesseract OCR, EasyOCR, PyTesseract, NumPy, pandas, Pillow, Scikit-Image

- AI-powered document digitisation system commissioned by Hamriyah Free Zone Authority UAE — automating extraction of 40+ structured fields from Emirates ID cards (front & back), international passports, and visa documents via a 6-stage pipeline: Upload → YOLOv8 Detection → Region Cropping → Dual OCR (Tesseract + EasyOCR) → Data Cleaning → Django ORM storage.
- Deployed 4 pre-trained YOLOv8 object detection models (~188 MB total: best_front_image.pt, best_back_image.pt, passport.pt, visa.pt) for precise field-level region-of-interest localisation across all document types, with OpenCV-based dynamic margin cropping and biometric extraction (person photos, signatures).
- Built ICAO 9303-compliant MRZ (Machine Readable Zone) parsing engine for passports, plus custom data cleaning algorithms: multi-format date parsing with year disambiguation, nationality code → country name mapping, gender standardisation, and regex-based OCR error correction — reducing manual post-processing to near zero.
- Reduced document processing time from minutes to seconds per document with duplicate-prevention database constraints, Django Admin CMS for record management, and a responsive 8-template web interface with modal image viewers.

Palletfly – Enterprise B2B Supply Chain & Inventory Management | dbr.palletfly.com

Tech: Django 3.2, Django Channels, DRF, Huey, Redis, PostgreSQL, Wagtail CMS, AWS (RDS, S3), Docker, GitHub Actions, Google Sheets API, Amazon Product API, Sentry

- 40,000+ lines of Python across 10 Django apps, 91+ ORM models, 308+ views, 26+ DRF serializers, and 380+ schema migrations — managing the full deal-to-shipment lifecycle for B2B logistics clients.
- Real-time collaborative editing via Django Channels + WebSockets + Redis pub/sub: multiple users can simultaneously update inventory sheets with zero-conflict live synchronisation.
- Async task engine (Huey + Redis): background Google Sheets import/export with rollback support, Amazon ASIN price/inventory sync, and bulk email notifications — keeping the request cycle non-blocking.
- Separate DBR_API Django service sharing the production RDS PostgreSQL database, exposing token-authenticated REST endpoints for external client integrations without coupling to the main app.
- Production on AWS (RDS Lightsail + S3) with Sentry error tracking, django-simple-history audit trails on all models, and GitHub Actions pipelines (test / lint / coverage) enforcing quality gates on every commit.

Pricing Intelligence Platform – WMC Mortgage Comparison Engine

Tech: Django 5.0.2, Python 3.12, Google BigQuery, Redis, Gunicorn, Docker, Kubernetes (GKE), Google Artifact Registry, GitHub Actions, GCP Service Accounts

- Cloud-native fintech platform for the WMC (Wealth Management Comparison) Index — queries live mortgage pricing data from Google BigQuery across multiple UK lenders with 10+ filter parameters (bank, LTV %, interest rate range, mortgage type, booking fee, interest period) and dynamic sorting with pagination.
- Redis caching layer (django-redis, 5-min TTL via @cache_page) applied to BigQuery query results, significantly reducing GCP data warehouse query costs and improving response times under concurrent load.
- Kubernetes-native deployment on GKE (wmc namespace): Dockerfile → Google Artifact Registry → K8s manifest via envsubst templating → kubectl apply, with liveness (/health) and readiness (/readiness) probes for zero-downtime rolling updates.
- Fully automated GitHub Actions CI/CD: GCP service account auth → Docker build → Artifact Registry push (tagged by run number) → GKE credential fetch → K8s manifest deploy — triggered on every push to main.

TimeKeepers (WorkFly) – Multi-Tenant Workforce Management SaaS | [workfly.app](#)

Tech: Django 4.0, DRF, Django Channels, WebSockets, Celery, RabbitMQ, Redis, PostgreSQL, Angular 15, TypeScript, PrimeNG, AWS (ECS, RDS, S3, ECR, SES), Docker, GitHub Actions, Kotlin (Android)

- Production multi-tenant SaaS platform (workfly.app) for security guard companies — 8 Django apps, 5 user role types (Licensee, Guard, Customer, Admin, Vendor), 64+ Angular components, 39+ backend services, and a Kotlin Android companion app.
- Real-time engine built on Django Channels + WebSockets + Redis for live guard-management chat, GPS location streaming, and instant operational alerts.
- GPS-based geofencing time-clock system: validates guard clock-in/out within authorised location perimeters, auto-calculates weekly hours, and generates time sheets for payroll.
- Full async infrastructure: Celery + RabbitMQ for background report generation and email dispatch (AWS SES), with Redis caching and django-simple-history for complete audit trails.
- Fully automated AWS ECS deployment via GitHub Actions (Docker → ECR → ECS task update), with separate DEV and Production pipelines triggered by branch push and tagged releases respectively.

TRIXI TAXI – Real-Time Ride-Sharing Admin Dashboard

Tech: React 18, Redux Toolkit, Socket.io, WebSocket API, Google Maps API, Material-UI, Firebase Storage, Chart.js, Recharts, Formik/Yup, Axios

- Multi-role admin panel (Super Admin, Company Admin) for full taxi operation — drivers, vehicles, rides, companies, billing, and real-time tracking across 20+ screens.
- Dual real-time layer: native WebSocket for per-user live notifications and Socket.io for live driver GPS tracking with Google Maps marker updates and route visualisation.
- Automated fare calculation engine factoring distance (per km), duration (per minute), car type tier (Trixi/Electric/Premium/XL), and add-on charges — integrated with Google Maps Directions API.
- Billing module with PDF invoice generation (html2pdf.js), Excel export (XLSX), FCM push notifications to drivers, and multi-criteria transaction filtering.

Ericsson & Huawei LTE/WCDMA GPL Audit System – Telecom Network Configuration Auditing

Tech: Python 3.x, PostgreSQL (psycopg2), Apache Kafka, pandas, openpyxl, Jupyter Notebooks

- Automated Golden Parameter List (GPL) auditing across Ericsson and Huawei LTE (4G) and WCDMA (3G) networks — comparing live configurations in EricssonDataHub and HuaweiDataHub PostgreSQL instances

against 100+ golden parameter standards across multiple geographic markets (BTC, Panama, Jamaica, and others), reducing manual audit time by 80%.

- Implemented 5 audit execution modes — Standard, Unconditional, Conditional, Batch (large-scale multi-site), and Debug — each with schema validation (`check_table_exists` / `check_column_exists`), rollback-safe execution, and structured deviation CSV reports with site-level and market-level granularity.
- Integrated Apache Kafka producer/consumer pipeline (`kafkacode.py` / `kafka.py`) for real-time network parameter event streaming, enabling continuous compliance monitoring without batch scheduling delays.

EDUCATION

Bachelor of Science in Computer Science *June 2018*

University of Central Punjab (UCP), Lahore, Pakistan

CERTIFICATIONS

- The AI Engineer Course 2025: Complete AI Engineer Bootcamp — Udemy
- AI Engineer Agentic Track: The Complete Agent & MCP Course — Udemy
- n8n – AI Agents, AI Automations & AI Voice Agents — Udemy
- Python, Cloud & AI Certifications — Coursera (Machine Learning, Cloud Computing, Python)
- Full Stack Developer Certification — EVS Training Institute, Lahore